

Public Transport Data Analysis Tool

Degree Programme: BSc Computer Science

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Clients: N/A

Proprietary Interests: There are no proprietary interests or outside help arrangements for this project.

Other Promises: No other promises have been made to secure acceptance of this project.

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The project was proposed by Me (Mohammad Faisal)

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1.PROBLEM TO BE SOLVED

This tool addresses the difficulty non-technical users face in accessing Transport of London's historical data. Each year, organisations like Transport of London and London Datastore (Transport for London,2025) publish various datasets, including station entry and exit numbers, delays, and more. However, this data is often scattered across multiple tables and disorganised CSV or Excel files. The tool aims to streamline this process by cleaning up the clutter and presenting the data clearly and consistently - whether as a chart, graph, or a simple table with essential columns.

While existing software products like Google Maps are excellent for real-time planning, they do not provide historical analysis. Other tools like Microsoft Excel or professional Business Intelligence (BI) software are either too complex for non-technical users or cannot handle this volume of scattered data.

2.PROJECT OBJECTIVES

Main Objective:

The project shall develop a web application that lets users query and visualise historical data published by Transport for London (TfL).

The objective is considered met when a functional web tool is developed and deployed successfully, capable of ingesting scattered TFL data files and providing an interactive dashboard to visualise the data.

Sub-Objectives:

1. Database Design and Implementation

- **Objective:** This project shall design and implement a relational database (PostgreSQL) to efficiently store historical station usage data.
- The Test:
 - The database schema is documented with an Entity-Relationship Diagram (ERD).
 - The database is successfully built and is proven to store the source data files.

2. Data Ingestion Pipeline (ETL)

- **Objective:** This project shall build an 'ETL' (Extract, Transform, Load) pipeline in Python to process the raw, "scattered" TfL data files.
- The Test:
 - The Python script successfully reads all source CSV files.
 - The script cleans the data (e.g., standardises date formats, removes duplicates).
 - The script loads the cleaned data into the PostgreSQL database without error.

3. Web Application and User Interface

- **Objective:** This project shall build a dynamic web application with a user interface, fulfilling the mandatory design and build component.
- The Test:
 - The application loads in a web browser.

- The UI provides functional query options (e.g., a dropdown to select a station and a date picker).

4. Data Visualisation

- **Objective:** This project shall implement data visualisations within the web app to display the results of a user's query.
- The Test:
 - The application queries the database based on the user's input.
 - The application displays at least two accurate, interactive charts (e.g., a line graph showing trends over time).

3. Project Beneficiaries:

The beneficiaries of this project would be people who want to analyse historical public transport data as part of their job (for example journalist writing an article on how has the delay on a particular underground line in the last 5 years)

- **Urban Planning Students & Researchers:** These users need to understand long-term mobility patterns to model urban growth or test hypotheses. This tool will directly benefit them by providing a queryable database of multi-year trends, saving them weeks of manual data cleaning and allowing them to immediately analyse patterns in station usage or journey times.
- **Data Journalists** need to find and verify evidence-based stories in public data. This tool will benefit them by acting as an "insight finder," allowing them to quickly spot anomalies, compare station performance across different boroughs, and find data to support articles on transport network performance or inequality.
- **Commuters & Public Interest Groups** While not a real-time planner, this tool benefits interested commuters by providing historical context. They can explore how busy their home station *usually* is on a weekday, check if their commute has observably slowed over the last year, or see how event days impact local stations.

4. Work Plan

Deliverables: The Project Definition Document and the Final Project Submission.

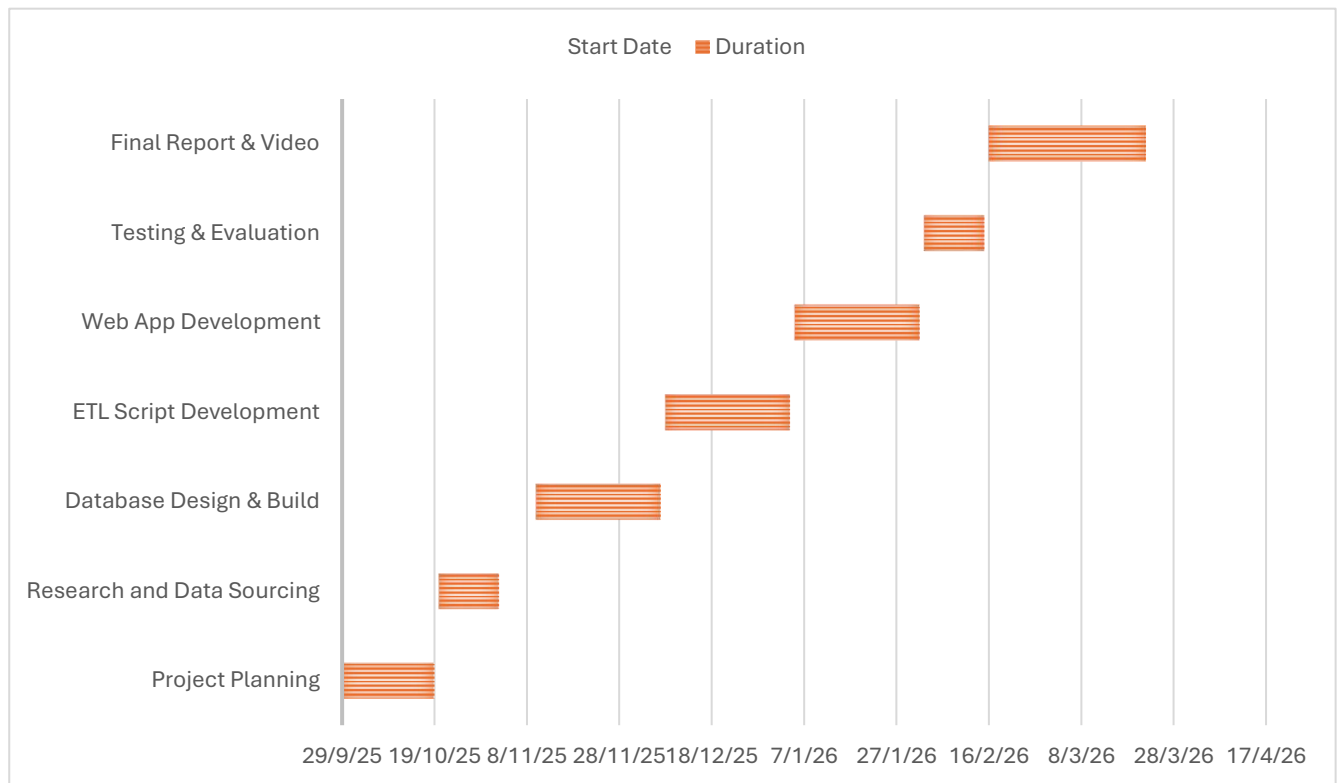
Project Outputs: The database, the final web application

- The database will be a PostgreSQL relational database.
- The ETL pipeline will be programmed in Python, using the Pandas library for data cleaning and SQL Alchemy to connect to the database.
- The web application will be built using Streamlit, a Python framework chosen to simplify development for a data-focused project.
- The primary data resource will be the historical "Tube passenger counts" and/or "Santander Cycle hire" datasets from the London Datastore.

The following table outlines the main activities, output, resources required and the planned timeline to control my progress.

Activity / Task	Outputs (What you will make)	Technology / Resources	Weeks
Project Planning	Approved PDD	MS Word, Project Handbook	1-3
Research & Data Sourcing	All TfL data files downloaded; tutorials for Streamlit & Pandas.	London Datastore, YouTube	4-6
Database Design & Build	ERD; a functioning PostgreSQL database with the correct tables.	PostgreSQL	6-8
ETL Script Development	The data un-scatterer: a working Python script.	Python, Pandas, SQLAlchemy	8-12
Web App Development	A working web app with a UI (dropdowns, date pickers) and charts.	Streamlit, Python	13-17
Testing & Evaluation	A list of test results proving the app's charts are accurate.	Test Plan, The App	18-19
Final Report & Video	The final report document and 15-minute demo video	MS Word, OBS Studio	15-22

Gantt Chart



5.Risks

Risk	Likelihood	Impact	Mitigation Strategy & Fall-back Plan
Data is too scattered and messy	High	High	Strategy: This is a core challenge, not just a risk. The work plan allocates a specific, dedicated block of 4 weeks (Weeks 13-16) <i>just</i> to this ETL script. Fall-back Plan: If the data is truly unmanageable, the project scope will be reduced to a single, simpler dataset (e.g., just Santander Cycles) instead of multiple, complex sources.

Legal risk of data usage.	Low	High	Strategy: Before any code is written, I will check the London Datastore's licensing terms for the TfL data. Fall-back Plan: If the data rights are too restrictive, I will pivot to a different public dataset (e.g., from Kaggle) that has a clear and open licence.
Learning new tech is difficult (Python/Pandas/Streamlit).	High	Medium	Strategy: The Research phase (Weeks 7-9) is allocated for learning these tools. I will study carefully by building small prototypes <i>before</i> working on the main project. Fall-back Plan: As the handbook suggests, I will drop a non-essential feature. The web app will be reduced to its minimal viable product: one dropdown and one chart. This satisfies the objective without complex features.
Data is too vast to process.	Medium	High	Strategy: The system will be designed to handle large data, but it will be <i>developed</i> using a smaller, manageable subset. Fall-back Plan: I will start with only <i>one</i> year of data to prove the system works. More years will be added <i>after</i> the core system is built, if time allows.

6. Legal, Social, Ethical and Professional Issues (LSEPI)

As defined in the Project Handbook, this project is a Category 1 project. The software product is a 'proof of concept' prototype. It is not intended for deployment or distribution to third parties, and I will not be responsible for its use.

However, as part of my professional duty, I have considered the LSEPI issues that *would* apply if this product were for actual use:

- **Legal:** The project relies on the TfL Open Data API and London Datastore. I must and will comply with all terms of service and data licensing agreements, including correct attribution.
- **Social:** The main social risk is data reliability. If my ETL script is wrong, the charts will be wrong, and a user could draw false conclusions. My final report will discuss these limitations and the importance of validation for any real-world use.
- **Ethical:** The data used is aggregated and anonymised, so the risk to individual privacy is low. However, I will treat the data with care to ensure no personal information is ever exposed.
- **Professional:** I will adhere to the BCS Code of Conduct. All re-used software (e.g., Python libraries) will be properly acknowledged. I will be honest about my project's limitations and will not claim it is a commercial-grade product

Declaration of Usage of AI:

As per the university's rules on generative AI, this appendix details my use of an AI assistant (Google's Gemini) in the preparation of this Project Definition Document.

My use was as a planning assistant and writing guide to help structure my ideas, not to generate the ideas themselves.

1. **Inputs:** I provided the AI with my core project idea (TfL Data Tool), my specific problems, and my personal worries. I also provided direct quotes from the Project Handbook for the AI to analyse.
 - a. **How it was used:** The AI helped me to: Structure my 'Project Objectives' to be 'testable' with 'Evaluation Criteria,' as required by the handbook. Turn my personal worries into a formal 'Risks' table with fall-back plans. Refine the wording of my 'LSEPI' section to correctly identify as a Category 1 project.
 - b. Solve a technical formatting issue in Excel for my Gantt chart.

2. **My Work:** I reviewed, edited, and rewrote all AI-assisted text to ensure it was factually correct and accurately reflected my own project plan.

References

1. Transport for London. (2025). *TfL Open Data*. [Online] Available at: <https://tfl.gov.uk/info-for/open-data-users/> (Accessed: 3 November 2025).

Research Ethics Review Form for BSc and MSci Projects

Computer Science Research Ethics Committee (CSREC)

<http://www.city.ac.uk/department-computer-science/research-ethics>

Undergraduate students undertaking their final project in the Department of Computer Science must consider the ethics of their project work and ensure that it complies with research ethics guidelines and the law for data protection. In some cases, a project will need approval from an ethics committee before it can proceed. Usually, but not always, this will be because the student is involving other people (“participants”) in the project.

To ensure that they give appropriate consideration to ethical issues, all students must complete this form and attach it to their project definition document (PDD). There are two parts:

PART A: Ethics Checklist. All students must complete this part.

The checklist identifies whether the project requires ethical approval and, if so, where to apply for approval.

PART B: Ethics Proportionate Review Form. Students who have answered “no” to all questions in A1, A2 and A3 and “yes” to question 4 in A4 in the ethics checklist must complete part B as well. The project supervisor or consultant has delegated authority to provide approval in such cases that are considered to involve MINIMAL risk. The approval may be **provisional** – *identifying the planned work with human end user participants as likely to involve MINIMAL RISK*. In such cases you must additionally seek **full approval** from the supervisor or consultant as the project progresses and details are established. You must obtain **full approval** in writing, before recruiting and engaging with human end users participants for your project.

A.1 If you answer YES to any of the questions in this block, your consultant/supervisor must have obtained approval for the project from an appropriate external ethics committee, and you need to have received written confirmation of this from him/her. Students cannot themselves apply for ethics approval in this case as the project is considered high risk". This type of research is not covered by City's process, and external approval from an appropriate institution is required.		Delete as appropriate
1.1	Does your research require approval from the National Research Ethics Service (NRES)?	NO
1.2	Will you recruit participants who are covered by the Mental Capacity Act 2005?	NO
1.3	Will you recruit any participants who are covered by the Criminal Justice System, for example, people on remand, prisoners and those on probation?	NO
A.2 If you answer YES to any of the questions in this block your consultant/supervisor must have obtained appropriate ethics committee approval		Delete as appropriate
2.1	Does your research involve participants who are unable to give informed consent? <i>For example, people who may have a degree of learning disability or mental health problem, that means they are unable to make an informed decision on their own behalf.</i>	NO
2.2	Is there a risk that your research might lead to disclosures from participants concerning their involvement in illegal activities?	NO
2.3	Is there a risk that obscene and or illegal material may need to be accessed for your research study (including online content and other material)?	NO

2.4	Does your project involve participants disclosing information about protected characteristics (as identified by the Equality Act 2010)? <i>For example: racial or ethnic origin; political opinions; religious beliefs; trade union membership; physical or mental health; sexual life; criminal offences and proceedings</i>	NO
2.5	Does your research involve you travelling to another country outside of the UK, where the Foreign & Commonwealth Office has issued a travel warning that affects the area in which you will study? <i>Please check the latest guidance from the FCO - http://www.fco.gov.uk/en/</i>	NO
2.6	Does your research involve invasive or intrusive procedures? <i>These may include, but are not limited to, electrical stimulation, heat, cold or bruising.</i>	NO
2.7	Does your research involve animals?	NO
2.8	Does your research involve the administration of drugs, placebos or other substances to study participants?	NO
A.3 If you answer YES to any of the questions in this block, then unless you are applying to an external ethics committee or the Senate Research Ethics Committee (SREC), you must apply for approval from the Computer Science Research Ethics Committee (CSREC) through Research Ethics Online - https://researchmanager.city.ac.uk/. Depending on the level of risk associated with your application, it may be referred to the Senate Research Ethics Committee (SREC).		<i>Delete as appropriate</i>
3.1	Does your research involve participants who are under the age of 18?	NO
3.2	Does your research involve adults who are vulnerable because of their social, psychological or medical circumstances (vulnerable adults)? <i>This includes adults with cognitive and / or learning disabilities, adults with physical disabilities and older people.</i>	NO
3.3	Are participants recruited because they are staff or students of City, University of London? <i>For example, students studying on a particular course or module. If yes, then approval is also required from the Head of Department or Programme Director.</i>	NO
3.4	Does your research involve intentional deception of participants?	NO
3.5	Does your research involve participants taking part without their informed consent?	NO

3.5	Is the risk posed to participants greater than that in normal working life?	NO
3.7	Is the risk posed to you, the researcher(s), greater than that in normal working life?	NO
A.4 If you answer YES to the following question and your answers to all other questions in sections A1, A2 and A3 are NO, then your project is deemed to be of MINIMAL RISK. If this is the case, then you can apply for approval through your supervisor under PROPORTIONATE REVIEW. You do so by completing PART B of this form. If you have answered NO to all questions on this form, then your project does not require ethical approval. You should submit and retain this form as evidence of this.		<i>Delete as appropriate</i>
4	Does your project involve human participants or their identifiable personal data? <i>For example, as interviewees, respondents to a survey or participants in testing.</i>	NO

PART B: Ethics Proportionate Review Form

If you answered YES to question 4 and NO to all other questions in sections A1, A2 and A3 in PART A of this form, then you may use PART B of this form to submit an application for a proportionate ethics review of your project. Your project supervisor has delegated authority to review and approve this application under proportionate review. You must receive final approval from your supervisor in writing before beginning the planned research.

However, if you cannot provide all the required attachments (see B.3) with your project proposal (e.g. because you have not yet written the consent forms, interview schedules etc), the approval from your supervisor will be **provisional**. You **must** submit the missing items to your supervisor for approval prior to commencing these parts of your project. Once again, you must receive written confirmation from your supervisor that any provisional approval has been superseded by with **full approval** of the planned activity as detailed in the full documents. **Failure to follow this procedure and demonstrate that final approval has been achieved may result in you failing the project module and/or result in an academic misconduct investigation.**

Your supervisor may ask you to submit a full ethics application through Research Ethics Online, for instance if they are unable to approve your application, if the level of risks associated with your project change, or if you need an approval letter from the CSREC for an external organisation.

B.1 The following questions must be answered fully. All grey instructions must be removed.		<i>Delete as appropriate</i>
1.1	Will you ensure that participants taking part in your project are fully informed about the purpose of the research?	NO
1.2	Will you ensure that participants taking part in your project are fully informed about the procedures affecting them or affecting any information collected about them, including information about how the data will be used, to whom it will be disclosed, and how long it will be kept?	NO
1.3	When people agree to participate in your project, will it be made clear to them that they may withdraw (i.e. not participate) at any time without any penalty?	NO
1.4	Will consent be obtained from the participants in your project?	NO

	Consent from participants MUST be obtained if you plan to involve them in your project or if you plan to use identifiable personal data from existing records. “Identifiable personal data” means data relating to a living person who might be identifiable if the record includes their name, username, student id, DNA, fingerprint, address, etc.	
1.5	Have you made arrangements to ensure that material and/or private information obtained from or about the participating individuals will remain confidential?	NO

B.2 If the answer to the following question (B2) is YES, you must provide details			<i>Delete as appropriate</i>
2	Will the research be conducted in the participant's home or other non-University location?	NO	
B.3 Attachments ALL of the following documents MUST be provided to supervisors if applicable. All must be considered prior to final approval by supervisors. A written record of final approval must be provided and retained.		YES	NO
Details on how safety will be assured in any non-University location, including risk assessment if required (see B2)			
Details of arrangements to ensure that material and/or private information obtained from or about the participating individuals will remain confidential (see B1.5) <i>Any personal data must be acquired, stored and made accessible in ways that are GDPR compliant.</i>			
Full protocol for any workshops or interviews**			
Participant information sheet(s)**			
Consent form(s)**			
Questionnaire(s)** <i>sharing a Qualtrics survey with your supervisor is recommended.</i>			
Topic guide(s) for interviews and focus groups**			
Permission from external organisations or Head of Department** <i>e.g. for recruitment of participants</i>			

***If these items are not available at the time of submitting your project proposal, then **provisional approval** can still be given, under the condition that you must submit the final versions of all items to your supervisor for approval at a later date. **All** such items **must** be seen and approved by your supervisor before the activity for which they are needed begins. Written evidence of **final approval** of your planned activity must be acquired from your supervisor before you commence.*

Changes

If your plans change and any aspects of your research that are documented in the approval process change as a consequence, then any approval acquired is invalid. If issues addressed in Part A (the checklist) are affected, then you must complete the approval process again and establish the kind of approval that is required. If issues addressed in Part B are affected, then you must forward updated documentation to your supervisor and have received written confirmation of approval of the revised activity before proceeding.

Templates for Consent and Information

You must use the templates provided by the University as the basis for your participant information sheets and consent forms. You **must** adapt them according to the needs of your project before you submit them for consideration.

Participant Information Sheets, Consent Forms and Protocols must be consistent. Please ensure that this is the case prior to seeking approval. Failure to do so will slow down the approval process.

We strongly recommend using Qualtrics to produce digital information sheets and consent forms.

Further Information

<https://www.city.ac.uk/about/governance/committees/cs-research-ethics>

<https://www.city.ac.uk/research/ethics/how-to-apply/participant-recruitment>

<https://www.city.ac.uk/research/ethics>